Group assignment

Causal ML

Setting

Groups: 1-4 students. Write us a mail as soon as possible but at latest 22.12., 14:00 stating

- either that you found a group yourself (one mail with group composition suffices)
- or that you need a group, then we will match you with other searching group members (please also state which task you prefer)

Submission: Upload your solution and slides until **19.1. 23:55** on ILIAS (one submission per group suffices)

Rules: Use whatever helps you making progress, but hand-in a group specific solution. For questions use the ILIAS forum.

Q&A: In case you start working on the group project over the christmas break and need help, Henri will offer a Q&A via Zoom on 2.1.2026. In case you want a Q&A meeting, please indicate this via mail by 1.1.2026.

Hint 1: Update the `grf` package to version 2.4.0. This should ensure that you receive the same results when using the same seeds.

Hint 2: Your computer may run out of memory for larger sample sizes. You may work with a random subsample of manageable size in this case.

Tasks

You can choose between the two tasks outlined below.

Task 1

In the lecture and tutorials, we have focused on methods that either assume the unconfoundedness assumption holds, or there is access to an instrumental variable. In practice however, there are other research designs to identify causal effects. In this task, you will use Double Machine Learning to estimate average treatment effects on the treated using Difference-in-Differences (DiD) under the "conditional parallel trends assumption." Along the course of this task, you can choose to either do this in a real application or in simulations. Subtasks 1.1 and 1.2 need to be done either way, in subtask 1.3, you can choose to work on 1.3a *or* 1.3b.

1.1 Refresh or kickstart your DiD knowledge (1P)

Depending on your background, you might have had varying degrees of exposure to the recent DiD literature. Your final submission should contain a brief introduction to the methodology and the underlying assumptions, so as a first step, get acquainted with how the recent DiD literature talks about causal effect estimation under (conditional) parallel trends. The lecture slides from the causal inference course in the summer (chapter: CI6_DiD), which you find on ILIAS, can serve as a starting point, although the introduction there focuses on the case without control variables. In addition, the homepage of Pedro Sant'Anna contains a lot of helpful material at different levels of depth, but you are of course free to find additional resources that advance your understanding.

1.2 Double ML for DiD using the Nuisance Parameters package (9 P)

In the lecture and tutorials, you have learned how to estimate various target parameters using Double Machine Learning. The estimation is based on Neyman-orthogonal score functions, which look different for different target parameters. Luckily for us, such scores also exist for average treatment effects on the treated. You can find them e.g., in the Documentation of the DoubleML package (the panel data version with "score = 'observational'). For additional background on these scores you can consult the papers by Sant'Anna and Zhao (2020) or Zimmert (2020).

Your task is to implement the estimation of the average treatment effect on the treated based on Double ML. For the estimation of the nuisance parameters use the NuisanceParameters package that provides a lot of flexibility to estimate cross-fitted nuisance parameters. On ILIAS, you find a folder with vignettes that illustrate how to use it. The package is currently under development, so if you run into issues with it, open an issue on github.

The result of this subtask should be an easy-to-use function that takes as input the data and options on how to estimate the nuisance parameters, and returns estimates of the ATT, the standard error, the t-statistic and the p-value. In addition, write a summary method, that creates a nice summary table.

From this point on, there are two versions of this task. You may either choose to use your function on a real data set, or evaluate it in a simulation study.

1.3a Real data set

1.3a.1 Find a data set (5P)

Find a (recent) research paper applying DiD, where the data is publicly available and the setup fulfills the following criteria:

- binary treatment of one group at one point in time
- control variables (time-invariant)
- at least $N = 100$ observations

1.3a.2 Descriptives (2P)

Briefly describe the application and the data. Provide suitable descriptive statistics.

1.3a.3 Replication (4P)

Replicate the main results from the paper using your own functions. As a sanity check, also use the pre-implemented functions of the DoubleML package and check whether the results are similar.

1.3a.4 Reflection (3P)

In your specific application, critically think about whether the assumptions for DiD are fulfilled (with a particular focus on the conditional parallel trends assumption). How do the authors of the original study justify the assumption? Do this in detail, using subject-specific reasoning and potentially data. Be creative on how to justify or criticize the assumptions in your setting.

1.3a.5 Feedback (1P)

From what you have learnt during this group assignment: what are obstacles you have faced during the project? What are potential problems with DiD in general and/or DiD based on Double Machine Learning in particular? In addition, provide feedback on the usability of the Nuisance Parameters package.

1.3a - Output

- R notebook(s) (not R markdown) including your solutions.
- Slides for a 10min pitch of your solution

1.3a - Focus of 10min pitch

- What is your application about? (not more than 2 slides)
- What are your results? Emphasize potential differences to the original study.

Alternatively:

1.3b Simulation Study

1.3b.1 Construct DGPs (5P)

Construct the DGPs from the section "Simulation 1: panel data are available" of Sant'Anna and Zhao (2020).

Write a function that allows to draw data from each of the different DGPs respectively.

1.3b.2 DGP Descriptives (2P)

Describe the different DGPs. How do they ensure that the conditional parallel trends assumption is satisfied?

1.3b.3 Simulation Study (4P)

Run a simulation study, where you demonstrate the performance of your ATT estimator in the different DGPs. Document the bias, variance, RMSE and coverage rates. Do so for different sample sizes.

1.3b.4 Interpret you results (3P)

Interpret your results in detail. Does your estimator perform well or only under certain circumstances? As an extension, you may also make (slight) adjustments of the DGPs to illustrate certain points for this subtask.

1.3b.5 Feedback (1P)

From what you have learnt during this group assignment: what are obstacles you have faced during the project? What are potential problems with DiD in general and/or DiD based on Double Machine Learning in particular? In addition, provide feedback on the usability of the Nuisance Parameters package.

1.3b - Output

- R notebook(s) (not R markdown) including your solutions.
- Slides for a 10min pitch of your solution

1.3b - Focus of 10min pitch

- Introduce the DGPs
- What are your results and how can you make sense of them?

Task 2: Additional estimators in the OutcomeWeights package

This task builds on a recent research paper of Michael: "Treatment Effects as Weighted Outcomes", introducing a framework and applying it to some estimators and some applications (see as gentle intro also Bluesky thread and Linkedin post). However, there is a lot of room for extensions, making it a nice assignment.

The OutcomeWeights package is currently only usable for some of the target parameters that can be estimated by Double Machine Learning. In particular, the function dml_with_smoother covers PLR, PLR-IV and AIPW estimators of the ATE, ATT, ATU and LATE.

Your task is to extend the function to also provide outcome weights for the local average treatment effect on the treated (LATT):

$$\tau_{LATT} = E[Y(1) - Y(0) \mid D(0) = 0, D(1) = 1, D = 1]$$

as well as the local average treatment effect on the untreated (LATU):

$$\tau_{LATU} = E[Y(1) - Y(0) \mid D(0) = 0, D(1) = 1, D = 0].$$

The LATT is equivalent to the ratio of the ATT of the instrument on the outcome and the ATT of the instrument on the treatment:

$$\tau_{LATT} = \frac{E[Y(1, D(1)) - Y(0, D(0)) \mid D = 1]}{E[D(1) - D(0) \mid D = 1]} \tag{1}$$

Similarly, the LATU is:

$$\tau_{LATU} = \frac{E[Y(1, D(1)) - Y(0, D(0)) \mid D = 0]}{E[D(1) - D(0) \mid D = 0]}. \tag{2}$$

Hint: In this paper, the treatment is denoted by D instead of W. Apart from that, everything should look familiar.

Spirit: Given that we know the general steps but did not go through all of them in detail, there might be unexpected obstacles. In case you run into such problems, we will provide feedback sessions on demand.

2.1 Familiarize yourself with the material (2P)

To get familiar with the LATT, check this paper. Provide a short literature review on where the LATT (and potentially the LATU) are defined and used.

Study the `Task 2 starting point 2025.R` script and the paper on outcome weights, in particular section 3.2, to learn about the strategies involved in rewriting treatment effect estimators as weighted outcomes.

2.2 Identification of the LATT/LATU (2P)

Under the standard assumptions for instrumental variables: show that τ_{LATT} and τ_{LATU} are equal to the expressions in (1) and (2), respectively. Then, show identification of τ_{LATT} and τ_{LATU}.

Hint: The identification can be done for the numerator and denominator separately.

2.3 Influence Function (3P)

You have shown that τ_{LATT} and τ_{LATU} are two ratios of ATTs (or ATUs). Derive an expression for the influence function of the new target parameters using the chain rule for influence functions. How would you estimate τ_{LATT} and τ_{LATU}?

Hint: The Neyman orthogonal score for the ATT is given on the lecture slides.

2.4 Weight representation (3P)

In the package and in this exercise, we use smoothers for estimating the nuisance parameters. That means that the vector of z-specific predictions for Y_i can be expressed as:

$$\hat{\mathbf{Y}}_{\mathbf{0}}^{\mathbf{z}} = \mathbf{S}_{\mathbf{0}}^{\mathbf{z}}\mathbf{Y}, \quad \hat{\mathbf{Y}}_{\mathbf{1}}^{\mathbf{z}} = \mathbf{S}_{\mathbf{1}}^{\mathbf{z}}\mathbf{Y}$$

For $\hat{\tau}_{LATT}$ and $\hat{\tau}_{LATU}$, derive expressions for the outcome weights as done for the Wald-AIPW in equation (20) in the paper.

2.5 Extend the function (10P)

Write an extended version of the dml_with_smoother function. Your new function should add code starting from line 159 in the original function and enable the user to extract the outcome weights for the additional estimators "Wald_AIPW_LATT" and "Wald_AIPW_LATU" by adding code in get_outcome_weights method starting in line 325.

For this, create a fork of the OutcomeWeights package, in which you then alter the dml_with_smoother function and the get_outcome_weights method. For submission, copy and paste your new dml_with_smoother function and the new get_outcome_weights method into a new R script.

2.6 Replication (4P)

Replicate the balancing check of the average effects R notebook with the newly obtained outcome weights and quickly comment on the findings, comparing them with the original notebook. Provide summary statistics on the weights you obtained for the LATT and the LATU. Briefly comment.

2.7 User feedback (1P)

You have used the OutcomeWeights package and read at least parts of the paper. Share here any suggestions for improvements.

Output

- R script containing your new dml_with_smoother function, the get_outcome_weights method and additional functions, if needed.
- R notebook including your replication
- Slides for the 5min pitch of your solution

Focus of 5min pitch

- What were the challenging parts and how did you solve them?
- What are your results?